A brief description of the final qualifying work

Title: Optical Skin Diagnostic System with Artificial Intelligence

This final qualifying work is devoted to the creation of a noninvasive optical system for diagnosing the skin using artificial intelligence methods. The relevance of the study stems from the increasing prevalence of skin diseases – both oncological and chronic inflammatory – and the limitations of traditional methods (visual examination, dermatoscopy), which do not provide information about subsurface changes. The goal is to develop a unified platform that combines visual data (RGB images) with infrared scanning to reveal clinically hidden signs of pathology.

The core idea is a multimodal analysis: simultaneously capturing images from a standard webcam and data from an infrared sensor operating at an 880 nm wavelength, which makes it possible to assess both surface morphological features (color, texture) and subsurface processes (changes in microcirculation, inflammation). The hardware includes a Full HD webcam, a dedicated point-infrared sensor controlled by a microcontroller, and mechanisms to maintain a fixed distance to the skin while minimizing external interference. All processing is performed in MATLAB and comprises modules for image preprocessing, segmentation algorithms, neural network models for classifying pathologies, and LSTM networks to predict dynamic development.

When processing RGB images, lighting correction, artifact removal (reflections, hair), and region-of-interest segmentation are applied. Extracted features are then passed to a pretrained EfficientNet-B0 convolutional neural network, fine-tuned on dermatological datasets. Infrared data are calibrated, filtered for noise, and interpolated from discrete measurements into a thermal map of subsurface properties. Multimodal features are merged by concatenating the output vectors from the last hidden layer of the CNN with averaged infrared intensity values; this combined vector is fed to a fully connected neural network for final classification. In addition, an LSTM model has been developed that, based on a short time series of lesion area changes and infrared reflection parameters, can predict the risk of disease progression – for example, the transition from actinic keratosis to squamous cell carcinoma.

The user interface, created with MATLAB App Designer, allows real-time display of the captured RGB frame, visualization of the segmented lesion, overlay of the thermal map on the visible image, selection of the classification algorithm, and viewing of prognostic dynamics graphs generated by the LSTM. All collected data are automatically saved to a spreadsheet for further analysis and validation.

In the experimental section, dozens of patients with various diagnoses (melanoma, benign nevi, psoriasis, eczema, rosacea) were examined. For each patient, RGB images of the affected areas were taken from multiple angles and a series of infrared measurements were recorded in a grid at a fixed distance, accompanied by histological reports and clinical evaluations by a dermatologist. Comparing classification accuracy using only RGB data, only infrared data, and the multimodal model revealed a clear synergistic effect: the multimodal system achieved higher sensitivity and specificity than single-mode approaches. Predictions of psoriasis flare-ups and progression of actinic keratosis were made with acceptable accuracy, confirming the promise of preventive analysis.

The conclusion emphasizes that the proposed system – built from readily available components (webcam and infrared sensor) and MATLAB algorithms – demonstrates high diagnostic accuracy and ease of use in clinical practice. This approach can reduce unnecessary biopsies, accelerate diagnosis, and enable dynamic patient monitoring. Future research directions include expanding the measurement spectrum (adding UV detection), migrating computations to embedded devices (TinyML), and conducting multicenter clinical trials to adapt models for diverse skin phototypes.